LITERATURE REVIEW

Comparative and qualitative robustness for law-invariant risk measures

Volker Krätschmer, Alexander Schied and Henryk Zähle

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ANALYTICAL ABSTRACT

The paper asks whether risk estimates built by plugging an empirical distribution into a law-invariant convex risk measure can be called robust, given that Cont, Deguest and Scandolo (2010) used Hampel's classical notion to conclude that coherent measures such as Average Value at Risk are not robust while Value at Risk is. The authors argue that Hampel's weak topology forces insensitivity to the tails, which is exactly what risk measurement should avoid. They replace it with a refined notion built on a finer psi-weak topology, defined on Orlicz hearts, and show that the Delta-2 condition on the Young function is necessary and sufficient for continuity and robustness of the risk functional. They prove strong consistency of the plug-in estimator for stationary ergodic data, introduce a comparative order and an index of qualitative robustness, and show that the index equals the reciprocal of the smallest moment order on which the measure is finite. For distortion measures the index is computed from the integrability of the derivative of the distortion function, so that Average Value at Risk has index one. The contribution is purely theoretical and asymptotic: no finite-sample bound, dependence structure or empirical calibration is supplied.

1 Introduction

1.1 Paper identification and classification

The paper is a theoretical contribution at the border of mathematical finance and robust statistics. Its field is risk management and its subfield is the statistical estimation of law-invariant monetary risk measures. The full text was read from the arXiv version 4 (23 pages, including the proofs of all theorems and an appendix on the weak topology). The journal version appeared in Finance and Stochastics in 2014; the review follows the arXiv text, and journal-only changes, if any, were not inspected.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

Whether the plug-in estimator of a risk measure, computed from historical or simulated data, has a stability property that is meaningful for risk management, and which risk measures possess it. The deeper aim is to replace a binary verdict (robust or not robust) by a graded one.

WHY DOES THE QUESTION MATTER?

Capital allocated from a risk estimate is costly to revise. If the estimate moves erratically when the sample is refreshed, capital requirements oscillate. At the same time, a risk measure that ignores the tail cannot warn of tail events. The two requirements pull in opposite directions, and the paper formalises the trade-off.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Cont, Deguest and Scandolo (2010) showed that Hampel's qualitative robustness is, by Hampel's theorem, equivalent to continuity of the risk functional in the weak topology, and that no law-invariant coherent risk measure (not even the expectation) satisfies it, whereas Value at Risk essentially does. What was missing is a notion of robustness that is not blind to tails and that can rank risk measures rather than split them into two classes.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes robustness with respect to a finer metric, the Prohorov psi-metric, on uniformly psi-integrating sets of laws. It works on Orlicz hearts, proves a Hampel-type theorem and its converse for the psi-weak topology, proves a Skorohod representation in Orlicz space, and obtains a characterisation by the Delta-2 condition. It defines an index of qualitative robustness with values in the unit interval and computes it for distortion risk measures. The central finding is that robustness is a continuum governed by the highest moment on which the risk measure is finite.

2 Research Problem and Hypothesis

2.1 Research question

PAPER Can the plug-in estimator $\hat{\rho}_n = R_\rho(\hat{m}_n)$ of a law-invariant convex risk measure be shown to be continuous, consistent and qualitatively robust once the weak topology is replaced by a topology that controls tail integrability, and can the resulting degree of robustness be measured by a number that orders risk measures?

2.2 Hypotheses

- The Hampel verdict against coherent risk measures is an artefact of the weak topology, and robustness is lost only partially, not entirely, when Value at Risk is replaced by Average Value at Risk.
- Continuity of R_ρ in the Orlicz-induced Ψ-weak topology holds for every convex risk measure on the heart H^Ψ exactly when Ψ satisfies the Δ_2-condition.
- A single index, taking values in $[0, 1]$ for convex measures, measures how much tail sensitivity a risk measure keeps and can be computed for distortion measures.

2.3 Existing literature and research gap

PAPER The authors build on Cont et al. (2010) for the robustness question, on Cheridito and Li (2009) for Orlicz hearts as the natural domain of convex risk measures with unbounded losses, on Krätschmer, Schied and Zähle (2012) for the general theory of qualitative robustness of tail-dependent functionals, and on Hampel (1971) and Huber and Ronchetti (2009) for the classical theory. CRITIQUE The gap is real: the weak topology treats a compactly supported law and a heavy-tailed law as arbitrarily close, so a risk measure that is continuous for it must ignore tails.

2.4 Claimed contribution

CRITIQUE The paper claims a refined notion of robustness, a comparative order, an index, consistency, continuity, a Skorohod representation and explicit indices for distortion measures. All of these are proved. What is not demonstrated is that the new notion is the right one for practice: the paper offers no calibration, no simulation and no comparison with a finite-sample criterion, so the claim that the index captures the practical robustness and sensitivity trade-off rests on interpretation rather than evidence.

3 Theoretical Framework and Mechanism

3.1 Setting, risk measures and the risk functional

PAPER Let $(\Omega, \mathcal{F}, P)$ be atomless and X a vector space of random variables containing the constants, interpreted as profit and loss (P&L). A map $\rho : X \to \mathbb{R}$ is a convex risk measure if it is monotone, convex and cash additive:

$$\begin{aligned} X \le Y \Rightarrow \rho(X) \ge \rho(Y), \\ \rho(\lambda X + (1-\lambda)Y) \le \lambda\rho(X) + (1-\lambda)\rho(Y), \\ \rho(X+m) = \rho(X) - m. \end{aligned} \tag{1}$$

Monotonicity says that a uniformly better position is less risky. Convexity says that diversification does not increase risk. Cash additivity says that adding m units of cash lowers the capital requirement by exactly m. Law invariance, $\rho(X) = \rho(\tilde{X})$ whenever X and $\tilde{X}$ have the same law, is equivalent to the existence of a risk functional on laws,

$$\rho(X) = R_\rho(P \circ X^{-1}), \qquad \hat{\rho}_n = R_\rho(\hat{m}_n), \qquad \hat{m}_n = \frac{1}{n}\sum_{i=1}^{n} \delta_{X_i}. \tag{2}$$

R_ρ Risk functional mapping a probability law on the real line to the risk of any position with that law.

$\hat{m}_n$ Empirical distribution of n historical or simulated observations $X_1, \dots, X_n$.

$\hat{\rho}_n$ Plug-in estimator. Its sampling behaviour is the object of the paper.

Equation (2) makes the estimation problem one of continuity: if $\hat{m}_n \to \mu$ in some topology and R_ρ is continuous in that topology, then $\hat{\rho}_n \to R_\rho(\mu)$. The choice of topology is therefore the whole problem. For $\rho(X) = -E[X]$ the functional $R_\rho(\mu) = -\int x\,\mu(dx)$ is not continuous in the weak topology, since a mass of $1/n$ moved to a point at distance n changes the mean by one unit while the law moves by $1/n$ in the Lévy metric.

3.2 Orlicz hearts and the Delta-2 condition

PAPER For unbounded losses the natural domain is an Orlicz heart. A Young function $\Psi : \mathbb{R}_+ \to [0, \infty]$ is left-continuous, nondecreasing and convex with $\Psi(0) = 0$ and $\Psi(x) \to \infty$. Then

$$L^\Psi = \{X : E[\Psi(c|X|)] < \infty \text{ for some } c > 0\}, \qquad H^\Psi = \{X : E[\Psi(c|X|)] < \infty \text{ for all } c > 0\}, \tag{3}$$

with Luxemburg norm $\|X\|_\Psi = \inf\{\lambda > 0 : E[\Psi(|X|/\lambda)] \le 1\}$. Norm convergence has the useful characterisation $\|X_n - X_0\|_\Psi \to 0$ if and only if $E[\Psi(k|X_n - X_0|)] \to 0$ for every $k > 0$. The two spaces coincide if and only if

$$\exists\, C, x_0 > 0 : \ \Psi(2x) \le C\Psi(x) \quad \text{for all } x \ge x_0. \tag{4}$$

For $\Psi(x) = x^p/p$ the condition holds and $H^\Psi = L^\Psi = L^p$. For $\Psi(x) = e^x - 1$ it fails, which is why the entropic risk measure $\rho(X) = \beta^{-1} \log E[e^{-\beta X}]$ is finite on the heart but not on the whole space. The condition (4) says that doubling a large loss multiplies its Young penalty by at most a constant, that is, the tail penalty grows no faster than polynomially. It is the exact line between moment-type and exponential-type tail control.

3.3 The psi-weak topology and the Prohorov psi-metric

PAPER For a weight function ψ (continuous, $\psi \geq 1$ outside a compact set), the ψ-weak topology is the coarsest one making $\mu \mapsto \int f d\mu$ continuous for all continuous f with $\sup |f/(1+\psi)| < \infty$. By Lemma A.1, $\mu_n \to \mu$ ψ-weakly if and only if $\mu_n \to \mu$ weakly and $\int \psi \, d\mu_n \to \int \psi \, d\mu$. It is metrised by

$$d_\psi(\mu, \nu) = d_{\text{Proh}}(\mu, \nu) + \left| \int \psi \, d\mu - \int \psi \, d\nu \right|. \tag{5}$$

The first term is the weak distance, the second is the discrepancy in the expected tail weight. For $\psi_p = |x|^p/p$ the topology is that of the Wasserstein metric of order p (Remark 2.9, citing Villani 2003). Taking $\psi \equiv 1$ recovers Hampel. A larger ψ penalises tail deviations more, so the choice of ψ sets how much tail sensitivity is demanded.

3.4 Qualitative robustness and the index

PAPER A set N of laws is uniformly ψ-integrating if $\lim_{M \to \infty} \sup_{\nu \in N} \int_{\{\psi \geq M\}} \psi \, d\nu = 0$. A risk functional is ψ-robust on $\mathcal{M}$ if it is robust for d_ψ on the source side and d_{Proh} on the sampling-law side, on every uniformly ψ-integrating $N \subset \mathcal{M}$, in the sense

$$\forall \mu \in N, \ \varepsilon > 0 \ \exists \, \delta > 0, n_0 : \quad d_\psi(\mu, \nu) \leq \delta \implies d_{\text{Proh}}\big(P_\mu \circ \hat{\rho}_n^{-1}, P_\nu \circ \hat{\rho}_n^{-1}\big) \leq \varepsilon \quad (n \geq n_0). \tag{6}$$

Here $P_\mu = \mu^{\otimes \mathbb{N}}$, so that the data are independent and identically distributed with law μ. The restriction to uniformly ψ-integrating sets cannot be dropped: Remark 3.3 shows it is equivalent to a uniform weak law of large numbers for $\psi(X_i)$. The index is defined by

$$\begin{aligned} \operatorname{iqr}(\rho) &= \Big(\inf\{ p \in (0, \infty) : R_\rho \text{ is } \psi_p\text{-robust on } \mathcal{M}(L^\infty) \} \Big)^{-1} \\ &= \Big(\inf\{ p \geq 1 : \rho \text{ is finite on } L^p \} \Big)^{-1}. \end{aligned} \tag{7}$$

The second equality is Equation (2.16) of the paper and follows from Theorem 2.16. Proposition 2.14 shows that any ψ for which a convex risk measure can be robust has at least linear growth, hence $\operatorname{iqr}(\rho) \leq 1$, with the value one for measures finite on L^1 and values near zero for measures that need very high moments.

3.5 Main theorems and what each says

PAPER Theorem 2.6: if ρ is a law-invariant convex risk measure on H^Ψ and (X_i) is stationary and ergodic with law of $X \in H^\Psi$, then $\hat{\rho}_n \to \rho(X)$ almost surely. Theorem 2.8: for finite Ψ, every law-invariant convex risk measure on H^Ψ has a Ψ-weakly continuous R_ρ if and only if (4) holds. Theorem 2.10: with (4), the following are equivalent for ρ on L^∞: finiteness on H^Ψ; continuity of R_ρ on $\mathcal{M}(H^\Psi)$; continuity on $\mathcal{M}(L^\infty)$; and $\|X_n\|_\Psi \to 0$ implies $\rho(X_n) \to \rho(0)$. Theorem 2.15 and 2.16 transfer this to robustness, with Theorem 2.16 stating that Ψ-robustness on $\mathcal{M}_1^\Psi$ is equivalent to finiteness of ρ on H^Ψ. Theorem 3.2 is Hampel's theorem for the ψ-weak topology (continuity implies robustness) and Theorem 3.4 is its converse (robustness plus weak consistency implies continuity). Theorem 3.5 gives the Skorohod representation: Ψ-weak convergence of laws is equivalent to the existence of random variables with those laws converging in the Luxemburg norm, if and only if (4).

3.6 Derivation: why Delta-2 is necessary (Theorem 3.5, sketch of the key step)

PAPER Suppose (4) fails. On an atomless space, $H^\Psi \neq L^\Psi$, so there is $Y \geq 0$ with $E[\Psi(Y)] < \infty$ and $E[\Psi(2Y)] = \infty$. Choose a_n with $2E[\Psi(2(Y \wedge a_n))] \geq n + \Psi(4n)$ and set $X_n = (Y - n)^+ \wedge a_n$. Then $X_n \to 0$ almost surely and $E[\Psi(|X_n|)] \leq E[\Psi((Y-n)^+)] \to 0$ by dominated convergence, so $P \circ X_n^{-1} \to \delta_0$ Ψ-weakly. Yet with $\ell(x) = \Psi(4x^+)$ and the inequality $\ell(x - y) \geq 2\ell(x/2) - \ell(y)$ implied by convexity,

$$E[\Psi(4|X_n|)] \;\geq\; 2E\big[\Psi\big(2(Y \wedge (a_n + n))\big)\big] - \Psi(4n) \;\geq\; n \;\to\; \infty, \tag{8}$$

so $\|X_n\|_\Psi \not\to 0$, and since the Luxemburg norm depends only on the law, no representation with norm convergence exists. The step uses that the weak part of Ψ-weak convergence is cheap (small mass far away) while norm convergence needs every dilation $k > 0$ to be controlled; without (4) a small mass at a very high level breaks the latter.

3.7 Distortion risk measures

PAPER Given a nondecreasing g with $g(0) = 0, g(1) = 1$,

$$\begin{aligned} \rho_g(X) &= \int_{-\infty}^{0} g(F_X(y))\,dy - \int_{0}^{\infty} \big(1 - g(F_X(y))\big)\,dy \\ &= g(0+)\,\operatorname{ess\,sup}(-X) + \int_0^1 \mathrm{V@R}_t(X)\, g'_+(t)\,dt. \end{aligned} \tag{9}$$

F_X	Distribution function of the P&L X.
g	Distortion function. It is concave if and only if ρ_g is convex, and then coherent (Dellacherie and Schmeidler).
$\mathrm{V@R}_t$	Value at Risk at level t, $-\inf\{y : F_X(y) > t\}$.

The second form shows that ρ_g is a weighted average of Value at Risk across levels, with weight $g'_+(t)$; the weight near $t = 0$ is the loading on the worst outcomes. Proposition 2.21 states that, for continuous

concave g and finite Ψ, finiteness on H^{Ψ} is equivalent to $g'_+ \in L^{\Psi^*}[0,1]$ with $\Psi^*(y) = \sup_{x\geq 0}(xy - \Psi(x))$ the conjugate Young function, and

$$\operatorname{iqr}(\rho_g) = \frac{q^* - 1}{q^*}, \qquad q^* = \sup\Big\{q \geq 1 : \int_0^1 \big(g'_+(t)\big)^q \, dt < \infty\Big\}. \tag{10}$$

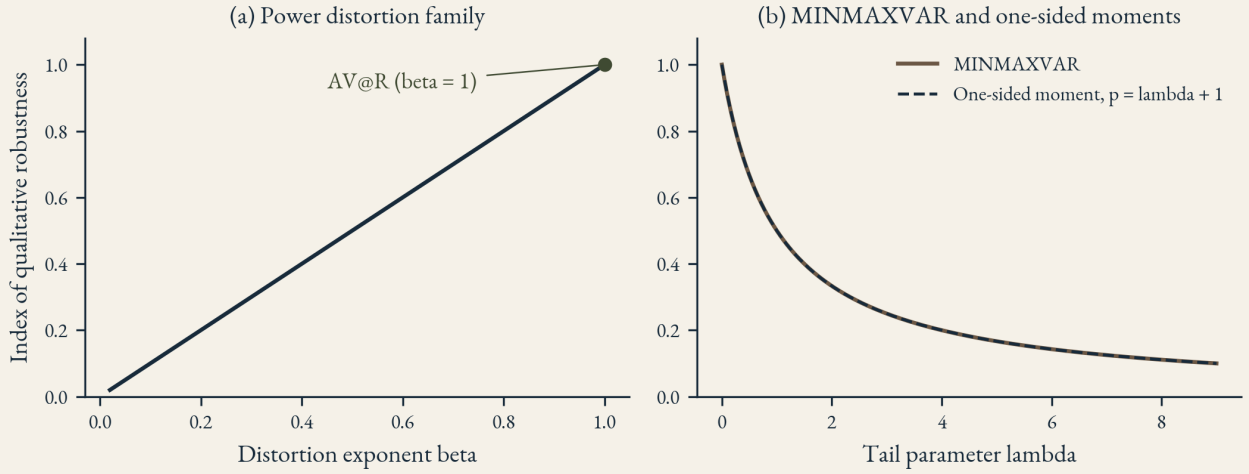
The proof uses Hölder's inequality in Orlicz spaces for the direction from (c) to (b), and Banach–Steinhaus with the Hardy–Littlewood inequality for the converse. The economic reading: the heavier the weight g'_+ places on the extreme tail (a singularity at $t = 0$), the higher the integrability exponent q at which g'_+ stops being integrable, and the lower the index. Examples: Average Value at Risk at level α has $g(t) = (t/\alpha) \wedge 1$, a bounded derivative, so $q^* = \infty$ and iqr $= 1$. For $g_\beta(t) = (t/\alpha)^\beta \wedge 1$ one has $g'_\beta \propto t^{\beta-1}$, in L^q iff $q(1-\beta) < 1$, so $q^* = 1/(1-\beta)$ and iqr $= \beta$. For MINMAXVAR, $g_{\lambda,\gamma}(t) = 1 - (1 - t^{1/(1+\lambda)})^{1+\gamma}$, $g' \sim c\, t^{-\lambda/(1+\lambda)}$ near zero and iqr $= 1/(1+\lambda)$. For the one-sided moment measure $\rho(X) = -E[X] + a\, E[((X - EX)^-)^p]^{1/p}$, iqr $= 1/p$.

Table 1. **Index of qualitative robustness for the risk measures treated in the paper**

Risk measure	Defining parameter	Index
Expected loss, Average Value at Risk	distortion with bounded derivative	1
Power distortion g_β	exponent $\beta \in (0,1]$	β
MINMAXVAR	tail parameter $\lambda \geq 0$	$1/(1+\lambda)$
One-sided moment measure	moment order $p \geq 1$	$1/p$

Source: official, Proposition 2.21 and Examples 2.20, 2.22 and 2.23 of Krätschmer, Schied and Zähle, reproduced by the reviewer.

Figure 1. **Index of qualitative robustness as a function of the tail parameter, closed forms of the paper**



Source: generated by the reviewer from the closed-form indices in Proposition 2.21 and Examples 2.20, 2.22 and 2.23. Dimensionless; no data involved.

4 Formal Setting and Proof Strategy

4.1 Absence of data

The paper contains no data, no simulation and no tables or figures of its own. The sections that the template devotes to data and empirical design are therefore replaced by the formal setting and the architecture of the proofs. The identifying variation of an empirical paper corresponds here to the mathematical device that makes the claims true: a stronger topology, plus a uniform law of large numbers on uniformly integrating sets.

4.2 Architecture of the proofs

Table 2. **Map of results, role and proof technique**

Result	Role	Technique
Theorem 2.6, consistency	Plug-in estimator converges almost surely	Birkhoff ergodic theorem, Skorohod representation, continuity of ρ in Luxemburg norm (Cheridito and Li 2009)
Theorems 2.8 and 3.5, Delta-2 characterisation	Sharp condition for continuity and for norm representation	Vitali theorem and uniform integrability; explicit counterexample through shortfall risk with loss $\Upsilon(8x^+)$
Theorem 2.10	Reduces finiteness on H^Υ to continuity at zero on L^∞	Lower semicontinuous extension to L^1 (Filipovic and Svindland), Cheridito and Li interior argument
Theorems 3.2 and 3.4, Hampel-type pair	Link robustness and continuity in the ψ-weak topology	Uniform Glivenko–Cantelli property, uniform law of large numbers (Chung 1951)
Proposition 2.14	Index is at most one	Counterexample $Y \geq 0$ with $E[\psi(Y)] < \infty$ and $E[Y] = \infty$
Proposition 2.21	Index of distortion measures	Hölder in Orlicz spaces, Banach–Steinhaus, Hardy–Littlewood

Source: reviewer's reading of Sections 2 to 4 and Appendix A of the paper.

4.3 Where does the identifying variation come from?

INFERENCE The result is driven by one construction, repeated in Theorems 2.8, 3.5 and Proposition 2.14: a sequence of laws that converge ψ-weakly to a point mass while a vanishing probability sits at a growing level, calibrated so that the risk measure (or the Luxemburg norm) of the sequence stays away from that of the limit. The positive results show that this construction is the only obstruction when the Young function satisfies the Δ_2-condition. The comparative content of the index is exactly the set of Young functions on which the measure stays finite.

4.4 Limits of the setting

- Laws live on the real line and positions are scalar P&L; multivariate or dynamic risk is not treated (the authors note that the framework of Krätschmer et al. 2012 allows multivariate extensions).
- Consistency requires stationarity and ergodicity, but the robustness definition (6) is stated for independent and identically distributed data.
- Cash additivity can be relaxed to cash coercivity (Remark 2.1) but the results are stated for the standard case.
- There is no rate of convergence: n_0 in (6) depends on μ and on ε.

5 Results

5.1 Consistency and continuity

EVIDENCE Theorem 2.6 establishes almost sure consistency of $\hat{\rho}_n$ for any law-invariant convex risk measure on H^Ψ under stationarity and ergodicity, which covers many GARCH processes, as the authors note. Theorem 2.8 shows this is not a triviality: with a Young function violating Δ_2, there is a convex risk measure (a shortfall risk with loss $\ell(x) = \Psi(8x^+)$) and a sequence of laws converging Ψ-weakly to δ_0 whose risks diverge, proved by showing that $z_n = \rho(X_n)$ solves $E[\ell(-X_n - z_n)] = 1$ and that a bounded z_n would contradict (8)-type growth. CRITIQUE The practical content is a warning: a consistent estimator can fail to be continuous in the data-generating law, so consistency from one estimating sequence need not transfer to another.

5.2 Robustness is a continuum

EVIDENCE Theorem 2.16 reduces Ψ-robustness of R_ρ on $\mathcal{M}_1^\Psi$ to finiteness of ρ on H^Ψ, which by Theorem 2.10 is continuity at zero of $\rho|_{L^\infty}$ in $\|\cdot\|_\Psi$. Consequently, Value at Risk and Expected Shortfall are not on opposite sides of a divide: Expected Shortfall is robust for ψ_1, that is, for laws close in the Wasserstein metric of order one, and Value at Risk is robust for every ψ but is not convex and therefore is outside the class for which the index is defined. INFERENCE The result reverses the policy implication of Cont et al. (2010): the Hampel failure of Average Value at Risk reflects that Hampel's topology is too coarse, not that the measure is unstable under reasonable perturbations.

5.3 Distortion measures

EVIDENCE The indices in Table 1 and Figure 1 show that distortion measures span the whole range $(0, 1]$. A power distortion with $\beta = 0.5$ has index one half, meaning robustness only with respect to ψ_2, that is, in Wasserstein distance of order two.

5.4 Illustrative calculation: contamination of a normal law

EXTENSION The paper has no numerical illustration, so the reviewer computes one, labelled as generated. Take the loss law $(1-\varepsilon)\,N(0,1)+\varepsilon\,\delta_M$ with $\varepsilon = 0.005$. The Prohorov distance to $N(0,1)$ is at most ε for every M. Value at Risk at 99 per cent has a quantile $q_N(0.99/0.995) = 2.574$, so it moves by 0.248 from 2.326 for every M. Average Value at Risk at 97.5 per cent equals

$$\mathrm{AV@R}_{0.975} = \frac{1}{0.025}\left[\int_{0.975}^{0.995} q_N\!\left(\tfrac{u}{0.995}\right) du + 0.005\,M\right], \tag{11}$$

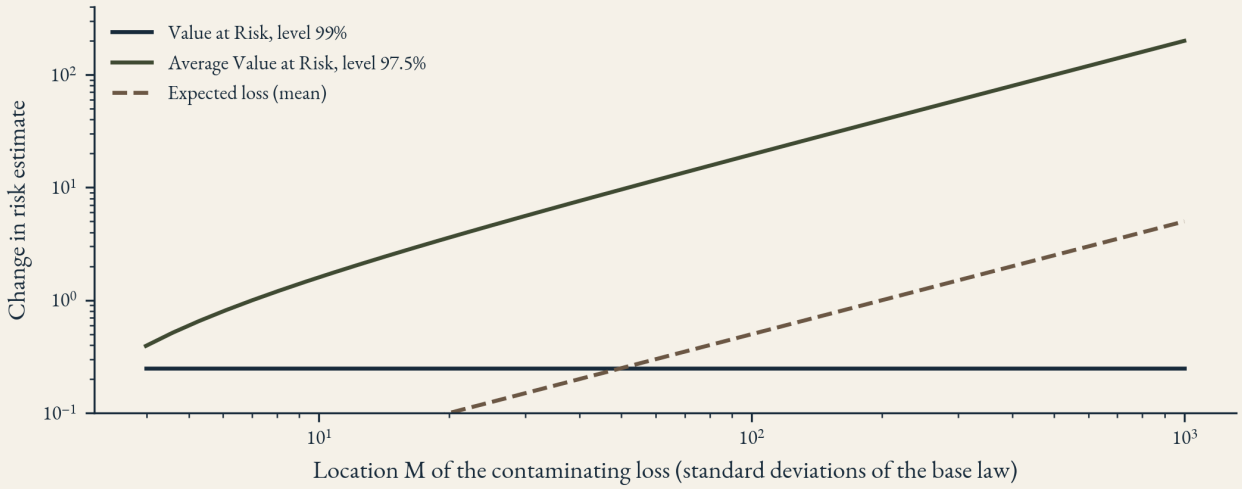
which is linear in M with slope $\varepsilon/(1-\alpha) = 0.2$, and the mean moves by εM. The weak-distance part of d_ψ stays at most ε, while the moment part $\left|\int |x|\,d\mu - \int |x|\,d\nu\right|$ grows like εM, the same linear rate as the mean and as the Average Value at Risk (slope 0.2 against 0.005 for the mean). The response of Average Value at Risk is therefore controlled by the ψ_1-distance, as the theory requires, and is unbounded for the weak distance alone. The Value at Risk response is bounded only because the contaminating mass 0.005 is smaller than the tail mass $1-0.99 = 0.01$.

Table 3. **Change in risk estimate under contamination of a standard normal loss law, mass 0.005 at level M**

Risk measure	M = 10	M = 100	M = 1000
Value at Risk, level 99 per cent	0.248	0.248	0.248
Average Value at Risk, level 97.5 per cent	1.597	19.597	199.597
Mean loss	0.050	0.500	5.000

Source: generated by the reviewer; exact quantile integration of the contaminated law, base values 2.326, 2.338 and 0.

Figure 2. **Change in three risk estimates as the contaminating loss M moves to the tail, contamination mass 0.005**



Source: generated by the reviewer. Base law N(0,1) for losses; vertical axis in loss units (standard deviations); both axes logarithmic.

EVIDENCE Reading Figure 2: the Value at Risk line is flat, the Average Value at Risk and mean lines are straight with unit slope on log scales. The qualitative separation that Cont et al. (2010) call robustness and sensitivity is visible, and so is the point of the paper: the linear growth of Average Value at Risk is exactly controlled by the first-moment term in d_ψ, so the measure is ψ_1-robust but not ψ_0-robust.

5.5 Tables and figures of the paper

The paper has none. Table 1 and Figures 1 and 2 are the reviewer's, and each has a stated origin.

6 Robustness and Additional Tests

Table 4. **Theoretical checks of the claims: concern, check, result and assessment**

Concern	Check in the paper	Result	Does it resolve the concern?
Is the Delta-2 assumption an artefact?	Theorem 2.8 and 3.5 give iff statements with explicit counterexample	Necessary and sufficient	Yes within the Orlicz framework
Is the restriction to uniformly integrating sets arbitrary?	Remark 3.3, equivalence with uniform weak law of large numbers	Cannot be dropped	Partly: justifies the definition, not that the class is the economically right neighbourhood
Do results rely on the Prohorov metric?	Remark: Levy metric would give a weaker notion; Prohorov extends to the multivariate case	Prohorov used	Partly: no check that the index is stable to the metric on the sampling side
Is cash additivity essential?	Remark 2.1, cash coercivity suffices	Results extend	Yes in principle, not proved in full
Is i.i.d. data too restrictive?	Consistency under stationarity and ergodicity	Robustness only i.i.d.	No: financial returns are dependent and heteroskedastic
Does the index discriminate between measures?	Examples for distortion measures	Whole range $(0, 1]$ covered	Partly: it only sees tail integrability, not the loading level

Source: reviewer's assessment of the checks reported in the paper.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Almost sure consistency of the plug-in estimator under stationarity and ergodicity (Theorem 2.6); the exact equivalence of continuity of R_ρ with the Δ_2-condition (Theorems 2.8, 3.5); the Hampel-type theorem and its converse for the ψ-weak topology; the formula (7) and the indices of Table 1. These are proved with complete arguments and checked by the reviewer against the printed proofs.

SUPPORTED BUT CONDITIONAL

That Average Value at Risk is robust. This holds for the ψ_1-metric and for sampling from uniformly integrating sets of independent observations; it does not extend to arbitrary heavy-tailed perturbations nor to dependent data.

SUGGESTIVE

That the index is the correct summary of the practical trade-off between robustness and tail sensitivity. It is a clean mathematical invariant but is not tied to any finite-sample or capital-stability metric.

NOT ESTABLISHED

Any quantitative statement about estimator stability at realistic sample sizes, any ranking of Value at Risk against Expected Shortfall in practice, and any claim about dependent data or about the multivariate case.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Replacement of a sharp dichotomy by a continuum, which defuses the argument against coherent measures drawn from Hampel's theorem.
- Results of independent interest: Skorohod representation in Orlicz space, continuity and consistency results.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The necessity direction is shown by explicit constructions, so the Delta-2 condition is a genuine boundary, not a sufficient technicality.
- Formula (7) turns robustness into a checkable integrability condition, which makes the index computable in one line for any distortion measure.
- The framework is general: it treats Value at Risk, shortfall, entropic and moment measures under one criterion, and the comparative order (Definition 2.17) is equivalent to inclusion of finiteness domains (Corollary 2.18).

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. By (7) the index is purely a moment-existence statistic: it records the smallest p for which ρ is finite on L^p. **Why it matters.** It cannot distinguish AV@R at level 0.5 from AV@R at level 0.001, which have identical index one but very different estimator variance, and it has no bearing on the speed at which $\hat{\rho}_n$ stabilises. A criterion intended to guide the choice of risk measure for capital allocation is silent on the quantity that drives capital instability. **Potential bias.** Overstating the practical difference between measures with different indices, and understating the difference between measures with equal index. **How it could be tested.** Compare the index with Monte Carlo variance and turnover of capital estimates across risk measures and levels, on heavy-tailed laws with known tail index.

OTHER WEAKNESSES

- Asymptotic statement with n_0 depending on μ and ε: no finite-sample guarantee. Test: derive or simulate concentration bounds for $\hat{\rho}_n$ under tail index a.
- Neighbourhoods are restricted to uniformly ψ-integrating sets. A heavy-tailed alternative with infinite ψ-moment is excluded by construction, though this is the threat that matters in a crisis. Test: examine estimator behaviour under alternatives outside the class.
- Independent data in the robustness definition. Volatility clustering inflates the variance of tail estimates; a version for strongly mixing or GARCH-type data is not given.
- Value at Risk, the measure that Hampel's notion favours, is outside the convex class, so the headline comparison between Value at Risk and coherent measures is not actually covered by the index, which the introduction suggests it is.
- No empirical calibration of the index, nor any statement on how a regulator or risk manager should choose ψ.

8 Contradictions and Edge Cases

CRITIQUE The result should disappear whenever tails are not controlled uniformly. For a Pareto-type loss with $P(L > x) \sim x^{-a}$ the quantile behaves as $q(t) \sim t^{-1/a}$ near zero, and for the power distortion g_β the integrand in (9) behaves as $t^{\beta-1-1/a}$, which is integrable at zero if and only if $\beta > 1/a$. The measure is therefore finite exactly when $a > 1/\beta$, in agreement with (7). For $\beta = 0.8$ a tail index above 1.25 is required, for $\beta = 0.5$ above 2, and for MINMAXVAR with $\lambda = 1$ above 2 again; these are illustrative calculations by the reviewer following from Proposition 2.21, not statements in the paper. Equity return tails with index near three to four are therefore within reach of $\beta \geq 0.5$, but operational-loss data with index near one are not.

INFERENCE At the opposite extreme, the entropic risk measure is finite on no L^p and therefore, by (7), has index zero, full tail sensitivity; the paper treats it as the standard example of a measure that lives only on a heart, but does not state its index.

CRITIQUE A second edge case is the choice of metric on the sampling side. Taking the Prohorov distance for d_B lets the sampling law of $\hat{\rho}_n$ move by large amounts in the far tail with small probability; for a risk manager who is concerned about the probability that the estimate is off by a fixed amount this is acceptable, but for one concerned about the expected size of capital revisions it is not. Third, with $\varepsilon = 0.005$ in Table 3 the 99 per cent Value at Risk is stable only because $\varepsilon < 0.01$. If the contamination mass were $\varepsilon = 0.02 > 0.01$ the 99 per cent quantile would equal M itself, and Value at Risk would be arbitrarily sensitive; this is the familiar discontinuity of quantiles in the tail mass and is invisible in the weak topology because the Prohorov distance stays at ε. The Hampel notion therefore also misclassifies Value at Risk in a regime where the tail mass exceeds the level.

9 Further Literature

Table 5. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Hampel (1971)	Qualitative robustness as continuity	Foundation, reinterpreted with a finer topology	Defines the notion the paper revises
Artzner, Delbaen, Eber and Heath (1999)	Coherent risk measures	Preceded; axioms	Source of the axioms behind Average Value at Risk
Föllmer and Schied (2002)	Convex risk measures	Preceded; framework	Convexity and the shortfall examples
Rockafellar and Uryasev (2002)	Conditional Value at Risk estimation	Applied different estimation view	Convex optimisation approach to the same measure
Acerbi and Tasche (2002)	Coherence of expected shortfall	Preceded; supports the measure	Why the measure is attractive despite Hampel
Cheridito and Li (2009)	Risk measures on Orlicz hearts	Foundation	Domain and norm continuity used throughout
Cont, Deguest and Scandolo (2010)	Robustness against sensitivity of risk estimators	Contradicted in conclusion	Source of the Hampel verdict that is revised
Krätschmer, Schied and Zähle (2012)	General qualitative robustness for tail-dependent functionals	Earlier paper by the same authors	Source of the index; this paper strengthens it
Ziegel (2016)	Coherence and elicitability	Different criterion	Elicitability gives another way of ranking measures
Emmer, Kratz and Tasche (2015)	Choice of risk measure in practice	Applied complement	Compares measures on practical grounds the index ignores

Source: first eight entries from the reference list and text of the paper (official); the last two are added by the reviewer from general knowledge and were not opened in this review.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** no finite-sample or rate statement. *Why it matters:* capital decisions are taken at fixed n. *Existing limitation:* n_0 is non-explicit. *Way forward:* concentration inequalities for L-statistics under tail index assumptions.
- **Gap:** dependent data. *Why it matters:* returns cluster in volatility. *Existing limitation:* robustness proved only for independent data. *Way forward:* mixing or GARCH-based sampling laws.

10.2 Literature-level gaps

- **Gap:** no link between qualitative robustness and the elicitability or backtestability of the measure. *Why it matters:* regulators evaluate measures by backtests. *Existing limitation:* the literatures use different criteria. *Way forward:* study whether robustness in the index sense is compatible with scoring-rule consistency.
- **Gap:** no empirical estimate of the index on real loss data. *Why it matters:* indices are only informative if data tails satisfy the moment condition. *Existing limitation:* tail index estimation is itself noisy. *Way forward:* confidence regions for the index based on extreme value estimators.

10.3 Methodological gaps

- **Gap:** a continuous sensitivity measure beyond the integrability index, such as an influence function in the ψ-weak topology. *Why it matters:* to distinguish levels within the same index class. *Existing limitation:* the index is integer-like in the moment sense. *Way forward:* infinitesimal robustness of tail-dependent functionals (the authors' earlier paper) applied to distortion measures.

11 Research Opportunities

Five ideas follow from the weaknesses identified above, each with the fields required for a serious project, then a ranking.

Idea 1. Finite-sample calibration of the index

RESEARCH QUESTION. Does the index predict the Monte Carlo stability of capital estimates across risk measures and tail indices?

MOTIVATION. The index is asymptotic and says nothing on variance.

HYPOTHESIS. Among measures with equal index, stability is monotone in the loading on the tail; among measures with different indices, the lower index has higher estimator dispersion at heavy tails.

ECONOMIC MECHANISM. Dispersion of $\hat{\rho}_n$ comes from the largest order statistics weighted by g'_+.

LITERATURE CONNECTION. Cont et al. (2010); Krätschmer et al. (2014).

DATA. Simulated Pareto and Student laws; daily equity and credit losses from public sources (to be sourced, not estimated here).

VARIABLES. Index, tail index, sample size, coefficient of variation and turnover of $\hat{\rho}_n$.

METHOD. Monte Carlo and block bootstrap.

IDENTIFICATION STRATEGY. Controlled variation of the tail index with known truth.

EXPECTED CONTRIBUTION. A bridge between the index and practical capital stability.

MAIN RISK. Results may show no systematic relation, which would itself be informative.

Idea 2. Robustness under dependence

RESEARCH QUESTION. Does the Delta-2 characterisation survive for strongly mixing or GARCH data?

MOTIVATION. Independence is the least realistic assumption of the paper.

HYPOTHESIS. Robustness holds on sets with uniform mixing rates, with the index unchanged.

ECONOMIC MECHANISM. Clustering concentrates extremes, increasing the effective variance.

LITERATURE CONNECTION. Boussama (1998) on GARCH ergodicity, cited by the authors; Zähle on strong laws.

DATA. Simulated GARCH and returns of liquid indices.

VARIABLES. Mixing coefficient, tail index, risk estimate.

METHOD. Uniform laws of large numbers for mixing sequences.

IDENTIFICATION STRATEGY. Theoretical, with simulations as sanity check.

EXPECTED CONTRIBUTION. Extends the results to the data that risk managers use.

MAIN RISK. Uniformity over mixing rates may fail.

Idea 3. Index versus elicitability and backtesting

RESEARCH QUESTION. Is there a trade-off between qualitative robustness and the existence of a consistent scoring function?

MOTIVATION. Expected Shortfall is not elicitable on its own, yet has index one.

HYPOTHESIS. Elicitable measures with the highest index are expectile-type measures.

ECONOMIC MECHANISM. Both properties concern continuity of the functional in the law.

LITERATURE CONNECTION. Ziegel (2016); Gneiting (2011).

DATA. Theoretical.

VARIABLES. Class of law-invariant convex measures, score functions.

METHOD. Characterisation through Orlicz-space duality.

IDENTIFICATION STRATEGY. Proof.

EXPECTED CONTRIBUTION. A unified criterion for choosing a regulatory measure.

MAIN RISK. Possibly a known result in disguise.

Idea 4. Estimating the index from data

RESEARCH QUESTION. Can the index of a given portfolio loss be estimated and tested?

MOTIVATION. The index is defined by finiteness on L^p, which depends on the data tail.

HYPOTHESIS. Using a Hill-type estimator one can build a confidence interval for the highest finite moment.

ECONOMIC MECHANISM. Tail index of losses determines the feasible set of measures.

LITERATURE CONNECTION. Extreme value theory; Proposition 2.21.

DATA. Public equity, credit and operational loss databases.

VARIABLES. Hill estimate, threshold, index bound.

METHOD. Extreme value inference with threshold selection.

IDENTIFICATION STRATEGY. Replication across subsamples and periods.

EXPECTED CONTRIBUTION. Operational rule to select a measure compatible with the data.

MAIN RISK. Threshold sensitivity.

Idea 5. Robustness for multivariate and dynamic risk

RESEARCH QUESTION. Do the ψ-weak results extend to risk measures of vector positions and to time-consistent dynamic risk?

MOTIVATION. Portfolio capital is a vector problem.

HYPOTHESIS. The Prohorov ψ-metric extends and the Δ_2 criterion carries over componentwise.

ECONOMIC MECHANISM. Aggregation of risk factors.

LITERATURE CONNECTION. Krätschmer et al. (2012) multivariate framework.

DATA. Theoretical, with simulated factor models.

VARIABLES. Vector P&L, aggregation functions.

METHOD. Orlicz hearts in several dimensions.

IDENTIFICATION STRATEGY. Proof.

EXPECTED CONTRIBUTION. Completes the programme.

MAIN RISK. Technical complexity without new insight.

11.1 Ranking

Table 6. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Finite-sample calibration	4	5	5	4	5	5	28
2. Dependence	4	3	4	4	5	4	24
3. Elicitability trade-off	5	3	5	4	4	4	25
4. Estimating the index	4	4	4	3	4	4	23
5. Multivariate and dynamic	3	2	4	4	4	3	20

Source: reviewer's subjective scoring.

The top three are ideas 1, 3 and 2. Idea 1 is the most feasible and answers the single most important weakness directly. Idea 3 is the most novel and links two strands of the literature that currently use different criteria. Idea 2 repairs the least realistic assumption.

12 Learning Framework

12.1 Prerequisite concepts

- Coherent and convex risk measures, law invariance, Value at Risk and Average Value at Risk.
- Weak convergence of measures, Prohorov and Lévy metrics, Skorohod representation.
- Hampel's qualitative robustness and the influence function.
- Orlicz spaces and hearts, Young functions, Luxemburg norm, conjugate Young functions.

12.2 Core concepts learned

- Robustness is topology-dependent; a finer topology buys tail sensitivity at the cost of robustness.
- The Δ_2-condition marks the boundary between polynomial and exponential tail control.
- Robustness of a convex risk measure is equivalent to finiteness on a moment space.

12.3 Important equations

Equation (2) for the plug-in estimator, Equation (5) for the Prohorov psi-metric, Equation (4) for the Δ_2-condition, Equation (7) for the index and Equation (10) for distortion measures.

12.4 Methods learned

Skorohod representation to move between laws and random variables; uniform laws of large numbers on uniformly integrating sets; Hölder and Banach–Steinhaus arguments in Orlicz spaces; counterexamples built from vanishing tail masses at growing levels.

12.5 Questions to test understanding

COMPREHENSION

1. Why is the expectation not weakly continuous, and what does that imply for Hampel robustness?
2. What does iqr equal for Average Value at Risk, and why?

ANALYSIS

1. Why does Theorem 2.8 need an atomless probability space?
2. Derive the index of g_β from Equation (10).

RESEARCH

1. Can a refined index separate Average Value at Risk at different levels?
2. What replaces uniform ψ-integrability for dependent data?

12.6 Papers to read next

Cont, Deguest and Scandolo (2010); Krätschmer, Schied and Zähle (2012); Cheridito and Li (2009); Ziegel (2016).

13 Conclusion

13.1 One-sentence thesis

Robustness of law-invariant convex risk measures is a continuum measured by the highest finite moment, and Average Value at Risk keeps the maximal degree compatible with tail sensitivity.

13.2 Three most important insights

- The Hampel verdict against coherent measures depends on a topology that ignores tails.
- The Δ_2-condition is exactly the boundary of continuity and robustness.
- Index equals the reciprocal of the smallest finite moment order.

13.3 Strongest evidence

The converse constructions in Theorems 2.8 and 3.5, which prove that the conditions are sharp.

13.4 Biggest weakness

The index measures only moment existence and is silent on finite-sample stability.

13.5 Most important research gap

Link between the index and finite-sample capital stability, and extension to dependent data.

13.6 Best research opportunity

Monte Carlo calibration of the index across tail indices and sample sizes.

13.7 Three papers to read next

Cont, Deguest and Scandolo (2010); Ziegel (2016); Emmer, Kratz and Tasche (2015).

14 References

Acerbi, C., and Tasche, D. (2002). On the coherence of expected shortfall. *Journal of Banking and Finance*, 26, 1487 to 1503.

Artzner, P., Delbaen, F., Eber, J.-M., and Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3), 203 to 228.

Cheridito, P., and Li, T. (2009). Risk measures on Orlicz hearts. *Mathematical Finance*, 19, 189 to 214.

Cont, R., Deguest, R., and Scandolo, G. (2010). Robustness and sensitivity analysis of risk measurement procedures. *Quantitative Finance*, 10, 593 to 606.

Emmer, S., Kratz, M., and Tasche, D. (2015). What is the best risk measure in practice? A comparison of standard measures. *Journal of Risk*, 18(2), 31 to 60.

Föllmer, H., and Schied, A. (2002). Convex measures of risk and trading constraints. *Finance and Stochastics*, 6, 429 to 447.

Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106, 746 to 762.

Hampel, F. R. (1971). A general qualitative definition of robustness. *Annals of Mathematical Statistics*, 42, 1887 to 1896.

Huber, P. J., and Ronchetti, E. M. (2009). *Robust Statistics*, second edition. Wiley, New York.

Krätschmer, V., Schied, A., and Zähle, H. (2012). Qualitative and infinitesimal robustness of tail-dependent statistical functionals. *Journal of Multivariate Analysis*, 103, 35 to 47.

Krätschmer, V., Schied, A., and Zähle, H. (2014). Comparative and qualitative robustness for law-invariant risk measures. *Finance and Stochastics*, 18(2), 271 to 295. Version read: arXiv:1204.2458v4, 17 July 2013.

Rockafellar, R. T., and Uryasev, S. (2002). Conditional value-at-risk for general loss distributions. *Journal of Banking and Finance*, 26, 1443 to 1471.

Villani, C. (2003). *Topics in Optimal Transportation*. American Mathematical Society, Providence.

Ziegel, J. F. (2016). Coherence and elicitability. *Mathematical Finance*, 26(4), 901 to 918.